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Physical Consequence, Not Localization, Governs Terrain-Induced Target Re-observability

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Abstract

Terrain-induced target re-observability is the problem a robot faces when a target it has already observed disappears behind terrain and must be made visible again by physical motion. This paper asks what information and what action reasoning that requires, and answers by elimination. We construct paired worlds that share terrain, pose, heading and a pixel-identical current observation yet require opposite re-observation actions, and in eight of eight pairs the successful action sets are disjoint, so no memoryless rule can solve both; a second family additionally shares the target memory and still requires different optimal actions. Over 240 (state, action) pairs, 43.1% of the actions a geometric visibility oracle endorses fail when executed, while the same oracle placed at the pose the robot actually reaches agrees with the physical outcome 97.9% of the time, so the missing quantity is the physical consequence of an action rather than its visibility. Testing that conclusion positively, on 2240 physically executed (state, action) pairs adding the candidate action to observation-plus-memory raises re-observation AUC from 0.854 to 0.934 whereas adding the current observation to the memory buys 0.002, and a planner built on the resulting action-conditioned consequence model raises RSR@10 from 0.250 to 0.792 and RSR@1 from 0.167 to 0.708 without privileged information. Three negative results bound how such a model may be used, and all three concern the prediction-to-decision step: perfectly localized anchor recovery still fails physically; a safety consequence head that is accurate and made better calibrated at the decision boundary yields exactly zero planning gain; and lookahead of two, three and five steps is not significantly better than one step on any budget while costing 1.8 to 2.1 times the motion. All results are in simulation with a vendored third-party locomotion policy; we claim the formulation, the benchmark, the failure analysis and the identification of the missing quantity, not a world model.

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1 Introduction

Robots that operate in natural terrain rarely enjoy a clear view of what they care about. A legged platform descending into a crater, a rover traversing a ridge line, or a manipulator reaching behind a boulder all face the same structural problem: the terrain itself determines what can be seen from where, and the robot's own motion is

what changes it. Partial observability in these settings is not sensor noise. It is a deterministic consequence of geometry, and it is therefore predictable in principle — but only if the robot reasons about how its next action will change what it can see.

This paper studies a specific and, we argue, under-analysed instance of that problem. The robot has already observed a target. At some later time the target is no longer visible — not because it moved or because the sensor failed, but because the robot's own position and the intervening terrain now block the line of sight. The robot must choose an action that makes the target visible again. We call this target re-observability, and it differs from the classical search problem in one decisive way: the robot has information about where the target is, and that information is still not enough.

The natural intuition is that moving toward a target makes it more likely to be seen. Terrain breaks that intuition. Consider a robot walking a straight line toward a target placed beyond a ridge. Measured along the traverse, the visibility sequence is not a single step from visible to hidden; it alternates:

stairs_up V X V X waves X V X V X

trench V X V X perlin X V X V X

pit V X V X basin X V X

barrier V X V X perlin_rough X V X V X

where V marks a position from which the target is visible and X one from which it is blocked. The number of transitions ranges from two to four on a single traverse. Two consequences follow. First, a policy that assumes monotonicity — ``get closer'' — is not merely imprecise; it is aimed at the wrong objective. Second, the moment of loss is not the moment of failure: a robot may lose the target and then re-acquire it purely as a side effect of continuing forward.

Existing lines of work each supply a necessary component but address a different

question. Lost-target search (Radmard and Croft, 2016; Radmard et al., 2018) maintains a belief over where the target went and selects search actions to reduce it; the mechanism is a belief and a search policy. Occlusion-aware active perception and next-best-view (OA-NBV authors, 2026; LEAP-NBV authors, 2026) select viewpoints by predicting visibility or information gain from a candidate pose, typically using geometry rather than execution. Object permanence filters (Peng et al., 2024) maintain a belief through periods of non-observation and act on the filtered estimate. Embodied visual tracking and target-loss recovery is an established research direction with its own benchmarks and systems (Wang et al., 2025; EVT-Bench authors, 2025); work in this line already combines action-conditioned world models with model-predictive control for recovering a lost target (CST-WM authors, 2026), and recent work explicitly couples visibility with reachability/feasibility in perception-aware design (authors, 2026). World-model planning more broadly (Yin et al., 2025; Wu et al., 2023; Robotic World Model authors, 2025) learns to predict the consequences of actions and plans with them.

None of this is unstudied, and we do not claim otherwise. What the literature lacks is a unified analysis: these lines are evaluated in their own settings, with their own observation and action interfaces and their own budgets, so the boundaries between them are never visible. In particular, no existing benchmark isolates the case in which the target is known and the failure is purely a matter of which physical action makes it visible again under terrain-induced occlusion. We place the same set of decision rules on a single task — terrain fixed, target static, action set identical, only the decision rule differing — and ask where and why each one breaks.

We do that here, and the answer is uncomfortable for all of them. We construct paired worlds: two scenes that share terrain, robot pose, heading and — we verify this by rendering the injected camera on the actual robot model — a

pixel-identical current observation, but in which the target was last seen from opposite sides. Across eight such pairs, the optimal re-observation action is opposite in the two worlds, and there is no action that succeeds in both. Since any function of the current observation is identical in the two worlds, no memoryless policy can solve both.

Knowing where the target is, however, is also not sufficient. A second family of paired worlds shares the current observation and the reconstructed target memory, and still requires different optimal actions. A rule that aims at the remembered target drives straight into the occluder; a heading-preservation rule holds a heading that was correct when the target was lost but is wrong after any lateral displacement. On 16 paired worlds, every classical decision rule scores 0/16 under a multi-step budget with a shared observation and action interface, while a privileged counterfactual oracle scores 13/16.

A third assumption also fails, and it is the one most likely to be silently wrong in practice. Next-best-view style methods predict the benefit of an action from a geometric re-projection of the believed target into a hypothetical camera pose. We implement that baseline with the strongest reasonable map — the exact height field — and measure over 240 (state, action) pairs that 43.1% of the actions geometry endorses fail physically. Crucially, the fault is not in the visibility model: placing the same oracle at the pose the robot actually reaches agrees with the physical outcome 97.9% of the time. What fails is the step before it — predicting where an action takes the robot. Because the discrepancy is a property of the closed loop rather than of the sensor, better perception cannot remove it.

Taken together, the three failures identify what a method for this problem must actually predict: the action-conditioned, physically-executed future observability $P(V_{t+H}=1 \mid O_t, M_t, a_{t:t+H})$, where V is measured after the action has been carried out.

We then test that conclusion positively rather than leaving it as a deduction. On 2240 physically executed (state, action) pairs, adding the candidate action to observation-plus-memory raises the re-observation AUC from 0.854 to 0.934, while adding the current observation to the memory buys 0.002. The information is in the action, not in seeing more. A planner built on the resulting consequence model — prediction of the executed pose followed by the already-validated visibility oracle — raises RSR@10 from 0.250 to 0.792 and RSR@1 from 0.167 to 0.708 with no privileged information, and is the only method we evaluate that chooses different actions in the two members of every paired world.

We also report where it still fails, and two negative results that constrain how any such model may be used. Its remaining errors are no longer action-selection errors but physical-termination errors (leaving the terrain, losing stability). And both an attempt to add safety to the decision rule and an attempt to plan over longer horizons fail in the same characteristic way: the prediction carries information, and the decision rule does not convert it.

This paper contributes:

A formulation of terrain-induced target re-observability as an action-selection problem, with the non-monotonicity of V along a traverse as its defining property, and an evaluation metric (RSR@K) whose implementation is audited against its own definition rather than assumed to match it (Section 2).

A procedurally generated benchmark of eight terrain-induced occlusion mechanisms, organised by physical mechanism and real-world analogue rather than as a difficulty ladder, built as exact box geometry so that the analytic visibility oracle and the MuJoCo collision surface are the same object; together with (i) paired worlds that differ in history and are provably identical in the current observation, (ii) a second set additionally identical in target memory, and (iii) a mechanism-support audit

that proves which mechanisms can express such an ambiguity and which structurally cannot.

A frozen observation/action interface under which every decision rule is evaluated on the same terrain, with the same budget, the same termination set and a policy-agnostic terrain contract, with all comparisons reported as paired bootstrap intervals rather than point estimates.

A failure analysis that separates the three assumptions, quantifies each, shows they are separable rather than three names for one failure, and identifies a failure class that the first version of our own taxonomy got wrong.

A measured answer to what the missing quantity is, and two negative results — in safety integration and in lookahead depth — showing that prediction quality is not the same thing as decision usefulness.

We are explicit about scope. All results are in simulation with a vendored third-party locomotion policy, on one controller; the oracle is an upper bound and not a deployable method; the learned consequence model is evaluated with terrain held out, which is generalisation within a mechanism family and not across mechanisms; and our target memory is a structured location memory rather than an open-vocabulary semantic one.

2 Materials and Methods

2.1 Problem formulation

Let O_t be the observation available at time t (pose, proprioception, a short action history and, where available, the camera), M_t the target memory — where and when the target was last seen — and a_t:t+H a sequence of physical actions. Write V_t+h in 0,1 for target visibility after executing that sequence. The task is equation

a^*_t:t+H \;=\; _a_t:t+H \; P\!(V_t+H=1 O_t, M_t, a_t:t+H).

equation

The target is lost when V_t=0 after having been observed, and

re-observed when there exists h <= H : V_t+h=1. Safety is deliberately

not part of this definition: the experiments below show that physical safety is

a dimension of the consequence model rather than part of the task statement, and mixing

it into Eq. 1 would confound the two.

2.2 The evaluation metric

The benchmark reports a difficulty curve rather than a single success flag, because the

quantity of interest is how many physical actions are needed to regain sight of the

target. For episode i and budget K,

equation

r_i(K) = 1\![\, h K : V_i,t+h = 1],

RSR@K = 1N_i=1^N r_i(K).

equation

Four policies may all eventually succeed while facing very different problems: one

action, or three, or more than the budget allows. Note what the definition deliberately

does not require: visibility to be held. Visibility under these mechanisms is

non-monotonic, and the task is to regain sight within a budget, not to keep it.

The implementation is audited against this definition rather than assumed to match it.

rsr_at evaluates Eq. 2 directly on the recorded per-step visibility

trace, while an episode also terminates at its first re-observation, so

success and steps_used <= K gives a second reading. A contract test asserts

the two agree for three terrains at every K = 1 6; if the termination semantics

ever change, the test fails instead of silently redefining the headline number.

Two companion metrics are reported alongside, and no single scalar combines them. The re-observation time T_reobs is the elapsed time from loss to the first re-observation, which separates ``wandered and then found'' from ``found immediately''. The action cost C_action is cumulative base-path length; it is averaged over every episode rather than only the successful ones, so that exhausting the budget does not look cheap, with the success-only figure reported alongside.

2.3 Benchmark: eight terrain-induced occlusion mechanisms

The suite is not ``eight kinds of terrain'' and it is not a set of obstacles built to break a baseline. Each entry is a terrain-induced target occlusion scenario: a situation in which a robot that has already seen a target loses sight of it and must regain sight by moving. Table 1 lists them, organised by physical mechanism rather than by surface name.

The mechanisms span five roughly orthogonal axes: occlusion direction, elevation dependence, spatial scale, visibility transition frequency, and terrain spatial complexity. Two points of framing are enforced mechanically rather than left to intent. First, the taxonomy is checked for the wording ``challenging / hard / difficult / adversarial terrain'' and refuses to render if it appears, because the difficulty is in the target's visibility, not in the driving. Second, perlin and perlin_rough are not claimed to be real surfaces: Perlin/fBm noise is used as a procedural representation of spatially correlated natural elevation variation, and the table gives the situation it stands in for.

All terrain is built as explicit box slabs rather than as a height field, so that the analytic visibility oracle and the MuJoCo collision surface are the same object. This matters: a height-field collider measurably stalls the locomotion policy, which would make the oracle and the physics disagree for reasons unrelated to the question.

2.4 Mechanism support validation

Observation-equivalent pairs with opposite optimal actions require the mechanism to be able to produce a one-sided winning set — every action that reveals the target moves the robot the same way laterally. We therefore test, per mechanism, the property that the construction requires, and report which mechanisms structurally cannot carry it. Table 2 gives the result; a high one-sided rate is necessary but not sufficient, and only window_wall has been searched end to end.

2.5 Frozen interface and baselines

Comparing decision rules across published work is normally confounded by each being evaluated with its own observation, its own action set and its own budget. We remove that confound by fixing one interface and expressing every baseline through it:

Environment (owns physics and camera, policy-agnostic) ->
Observation -> ReObservationPolicy.act(observation)
-> Environment.rollout(action) for privileged scoring only.

Three properties make the comparison meaningful. No privileged information appears in Observation: ground-truth visibility and the target's world position live in a separate PrivilegedState that only the evaluator and the oracle may read, and a contract test asserts this rather than trusting convention. The terrain is policy-agnostic: terrain modules must not import the API or any policy, enforced both by an AST scan of the terrain modules' imports and by a subprocess that imports each terrain module alone and fails if the policy stack comes with it. And counterfactuals do not perturb the episode: rollout saves and restores the integration state and resets the controller; without this the oracle's own probing would move the robot it is deciding for, a defect the contract test caught in the first implementation.

Every baseline implements the core decision mechanism of a line of work, not a full reproduction. Reactive always drives forward. HeadingLock holds

the heading recorded at the loss moment from IMU only. HistoryOnly aims the heading at the remembered target and drives. LostTargetSearch faces the belief and then closes on it. ObjectPermanence runs a constant-velocity belief filter and acts on the filtered estimate — kept as a filter rather than defined as equivalent to HistoryOnly, because on a static target the extrapolation term vanishes and the collapse is then a measured result rather than a stipulated one. NBV re-projects the believed target from a kinematic hypothetical pose and is deliberately given the exact height field, since an active-perception method is allowed a map and a failure should not be attributable to a weak one. Oracle physically rolls out every candidate and takes the fastest success; it is a privileged upper bound, never presented as a deployable method.

Fairness invariants, asserted by contract tests: every primitive costs the same simulated time, so one step is one unit of budget; the budget is actually granted, verified by forcing the visibility read to report ``hidden'' and checking that a failing episode consumes all N steps; one N, one termination set and one observation stream length for every policy; an action identifier outside the frozen set is an error and never a silent fallback; and RSR@N is monotone in N.

Statistics. Every proportion carries a Wilson score interval, not a normal approximation, because several results sit at or near 0 % or 100 % where the normal interval produces out-of-range bounds. Every comparison is paired: all policies run on the same terrains, so a difference in RSR is tested by resampling terrains with replacement and examining the distribution of the paired difference. Two degenerate cases are labelled rather than dressed up as statistics. Where two policies make outcome-identical choices on every terrain there is no variance to bootstrap and no p-value is reported, because reporting p=1 would present ``identical'' as ``not significant''. Where a difference distribution contains no

non-positive resamples the p-value is bounded as <5x10^-5 rather than printed as an exact zero.

2.6 Action-conditioned consequence model

The failures above identify the quantity to predict as Eq. 1. Rather than learning visibility end to end, we exploit the third result: since the visibility oracle is accurate at whatever pose the robot reaches, the model predicts the executed body-frame pose delta and the already-validated oracle is applied to the predicted pose. This keeps the validated part validated and localises the remaining error.

Training data is 24 terrains x 10 states x 10 actions = 2240 physically executed (state, action) pairs, positive rate 0.143. Features are the pose, the projected gravity vector, a short pose history, the target memory in body coordinates with an explicit presence flag, and the action. The memory block is body-relative and its age is carried, both of which must be maintained during any chained rollout. The targets are the executed pose delta and whether the target was re-observed. Splits are by terrain and never random, so a terrain family cannot appear on both sides.

The planner scores every candidate action with the model, takes the argmax, and replans after each executed step.

3 Results

3.1 H1: the current observation does not determine the action

We construct paired worlds that share terrain, robot pose, heading and a pixel-identical current observation, but in which the target was last seen from opposite sides. Across eight pairs: the rendered camera images are identical (mean and maximum absolute pixel difference both zero), the optimal re-observation action is opposite in the two worlds in 8/8, and the successful action sets are disjoint in 8/8.

Because any function of the current observation takes the same value in both worlds, no

memoryless policy can solve both. On 16 paired-history worlds the five classical rules score 0/16 within a ten-step budget.

This is not a statement that a particular baseline is weak. It is a statement that the task is structurally partially observable.

3.2 H2: a correct target memory is also not sufficient

A natural reply to H1 is to remember where the target is. We therefore construct a second family of pairs sharing the current observation and the reconstructed target memory, differing only in terrain hidden inside the shadow of a visible occluder — terrain that is invisible from the decision point and was never traversed.

Over six such pairs, the rendered observations are pixel-identical in 6/6, the target memory is identical in 6/6, and the optimal action differs in 4/6. Two properties must be reported precisely, because they bound the claim. First, the successful action sets overlap (0/6 disjoint), so H2 does not show that no fixed rule can succeed: what it shows is that a correct memory does not determine the optimal action. Second, on these pairs all six classical policies choose the same action (toward) and fail in both worlds, so H2 separates the baselines not at all — it is an information-level result, and presenting it as ``existing methods fail here" would be wrong.

Transferring the failure to the main suite is nevertheless unambiguous: on the 16 paired-history worlds, every classical rule scores 0/16, and a heading-preservation rule that holds the correct inertial heading fares exactly as well as a purely reactive one — paired difference 0.000 with interval [0,0], i.e.\ outcome-identical on every terrain.

3.3 H3: geometric visibility is not physical re-observability

For every candidate action at every decision state we compute two quantities. G(s,a) asks whether the target would be visible if the robot reached the ideal pose the action

nominally produces, using a kinematic displacement and the line-of-sight oracle — what an active-perception method assumes it can evaluate. P(s,a) asks whether the target is actually re-observed when the primitive is executed.

Over 24 terrains x 10 actions = 240 pairs, the confusion matrix is G=0,P=0=171, G=0,P=1=11, G=1,P=0=25, G=1,P=1=33. 43.1% of the actions geometry endorses fail physically. The disagreement is not one-sided — geometry also under-predicts in 4.6% of cases.

The mechanism is identified by the same experiment. Placing the same oracle at the pose the robot actually reaches makes it agree with the physical outcome in 97.9% of pairs. The pose prediction error is larger on the failing cell than on the agreeing cells by a paired bootstrap difference of +0.28 m with 95% interval [+0.11, +0.45] and p = 0.0005; decomposed, the error is 0.52 m forward, 0.34 m lateral, 0.40 m in height and 5.5 degrees of yaw.

The comparison of the point estimate is large, but the practical reading is what matters: no active-perception method is mis-implemented here. Candidate viewpoints scored by kinematic re-projection are simply a different quantity from where the robot ends up.

3.4 E4-1: the action carries the information that the observation does not

We ablate the input to the re-observation predictor with terrain held out (Table 3).

Adding the candidate action raises AUC by +0.080. Adding the current observation to the memory raises it by 0.002. The information is in the action, not in seeing more.

On the 16 paired-world decision states the O+M+a model reaches 0.8125 accuracy and switches its chosen action in 8 of 8 pairs — the thing a memoryless rule provably cannot do.

3.5 E4-2: a learned consequence model beats the analytic one, but not everywhere

The correct statement is that the learned model wins on the components that decide success, not that learning is better in all dimensions.

3.6 E4-3: consequence prediction converts into task performance

RSR@10 rises from 0.250 to 0.792 and RSR@1 from 0.167 to 0.708, with no privileged information. The paired comparison against HistoryOnly is +0.708 with a 95% interval of [+0.542, +0.875].

By contrast, no classical baseline significantly outperforms any other. Relative to HistoryOnly, Reactive, HeadingLock, LostTargetSearch and ObjectPermanence each differ by 0.000 with interval [0,0] — they succeed and fail on exactly the same terrain set. NBV differs by +0.042 with interval [0.00, 0.125] and p=0.356, so it cannot be claimed to be better. Only the privileged references separate.

3.7 E4-4: short history helps motion, not visibility

Visibility AUC is flat across every horizon, while pose error improves from 0.177 to 0.147 and saturates at four recent observations. History helps predict where the robot will end up, not whether the target will be visible — which is consistent with the localisation of the missing quantity in H3.

3.8 E4-5 and E4c: the safety paradox

Adding safety as a consequence head raises Oracle-S to 0.4107 against Oracle-V at 0.3884, so the safety term is worth having. The learned head is accurate, with ROC-AUC 0.900 and PR-AUC 0.606 on a positive rate of 0.083. Yet the planner gets worse at every penalty weight: 0.3661 without safety, 0.3125 with a hard veto, and 0.3438–0.3571 across soft penalties.

E4c separates the prediction from the rule. The head's overall calibration is healthy (ECE 0.042) while its calibration in the band |p-0.5| <= 0.15 — the only region where a threshold changes anything — is 0.197, a factor of 4.7. Temperature scaling

halves the decision-band ECE to 0.099. The planning improvement is zero. Across raw, temperature-scaled and isotonic variants, no operating point at any threshold or coverage beats not using safety at all, and every variant's optimum falls at coverage 0.

This is the more informative branch of the diagnostic: it rules out ``calibrate it better'' as the next step and locates the limit in the joint visibility–safety decision boundary rather than in the accuracy of the probability.

3.9 E4d: the horizon paradox

Against H=1, the paired success differences are -0.208 with interval [-0.458, +0.042], 0.000 with [-0.208, +0.208], and -0.125 with [-0.333, +0.083]. No horizon is significantly better on any budget, and every horizon costs 1.78–2.09 times the motion.

The per-terrain breakdown is what makes this informative rather than merely negative. The privileged reference genuinely requires multiple steps on six terrains. Lookahead does rescue two of them reliably — stairs_up and pit, solved by H >= 2 and not by H=1 — which confirms that re-observation is intrinsically sequential. But it breaks eight terrains that one step solves.

Model-error accumulation is not the explanation. The measured one-step pose error is lowest at H=2 (0.301 m) and yet H=2 performs worst. The mechanism is instead that the chained score _k p_k is monotonically optimistic in H: a longer horizon offers more opportunities to find a sequence that the model confidently believes will glimpse the target, and those are exactly the confidently wrong predictions. The one-step pose error (0.30–0.76 m) is of the same order as an action's displacement (0.5–1.8 m), so by the second or third chained step the predicted state is nearly uninformative.

3.10 E4e: a joint decision over one event does not rescue it either

E4c and E4d failed in the same shape, which suggests the fault is not in either head but in how they are combined. Both were serial compositions of quantities predicted independently: a visibility ranking with a safety term that vetoes it, and a per-step visibility score with a maximum taken over chained steps. A veto is the wrong shape — the planner does not want to know whether an action is unsafe, it wants to know whether the action achieves the goal, where the goal is re-observed and still on its feet. That is a single event.

We therefore predict that single event directly. A joint head is trained on P(V and safe | O,M,a) — the conjunction, not a product of two marginals, since multiplying marginals would reintroduce the independence assumption under test. Alongside it we test a conformal filter whose threshold is the out-of-fold quantile of the safe class's unsafe-score, which by construction bounds how often a genuinely safe action is discarded; this is the targeted fix for the measured failure that a fixed threshold on the safety score removes good and bad actions at similar rates.

Neither works. Against a baseline of 0.3661 one-step success, the joint head reaches 0.3527 (a gain of -0.013, intervening at 18.3% of states), and the best conformal filter reaches 0.3616 (-0.005). Combining them is worse still (0.3348–0.3571). The positive rates are 0.143 for visibility, 0.119 for visibility-and-safe and 0.083 for unsafe, so the conjunction is rarer than either component — and the privileged Oracle-S, which has ground truth, still improves on Oracle-V (0.4107 against 0.3884). Three successive principled fixes — hard and soft penalties, temperature and isotonic calibration, a joint head, and a conformal filter with a distributional guarantee — all fail to convert the safety information into planning improvement, while the same information with ground-truth labels improves it. We report this as the paper's open problem rather than as a solution: the gap is not in the accuracy of the predicted

quantities, and it is not closed by predicting the right quantity instead.

3.11 E-Odom: the task is not localisation recovery

The most damaging reading of this benchmark is that it reduces to state recovery: save the pose at which the target was last seen and use odometry to return to it. We implement three such baselines — steering back to the last-visible pose, dead-reckoned backtracking along the recorded walk-in, and returning to that pose before advancing again along the original sight ray — and evaluate them under three odometry conditions: perfect, moderate drift (2% range, 1^ per step heading) and realistic drift (5% range, 3^ per step).

Two results settle the objection. First, perfect odometry does not solve the task: the best anchor-recovery baseline reaches RSR@10 = 0.250, the same as the classical rules and far below the consequence planner's 0.792. Second, and more pointedly, six of its 24 failures under perfect localisation are physical terminations — the robot falls or leaves the terrain while returning to a pose it knows exactly. A perfect localization memory cannot guarantee physical re-observability. The drift conditions then make the point sharper rather than weaker. Perfect and moderate drift produce identical results, and realistic drift changes RSR@10 by 0.000, raising cost by 0.13 m and moving one failure from B to F. The failure is therefore not an odometry-accuracy problem: it is the physical-consequence gap of H3, reached from a completely independent direction.

3.12 Failure taxonomy

Classical policies predominantly fail because they select ineffective actions, whereas the learned consequence planner substantially reduces action-selection failures and instead exposes physical-termination failures. The failure type changes with the capability of the planner, which is itself evidence that the modelled quantity is the right one and that the remaining gap is coverage of the consequence model rather than

candidate diversity.

Two limitations of the taxonomy are stated rather than smoothed over. The classes C and D have one example each, which is enough to establish that the class occurs and not enough to report a rate. And the classification of F was a correction: our first version recorded falls and boundary exits as action-selection failures, which is an attribution error, since the action was chosen on a defensible prediction and the prediction of stability was what failed.

4 Discussion

4.1 Why the classical baselines are indistinguishable

Not because they are poor algorithms. On 24 terrains, five of the six produce outcome-identical success patterns, differing by 0.000 with interval [0,0]. They are not merely similar; they make the same decision.

The reason is structural. All five mechanisms operate on the same quantity — ``which way should I face or move" — that is, O_t, M_t -> a heading or a direction. None of them explicitly models the map O_t, M_t, a -> S_future. Since what is missing is the same thing in all of them, no amount of care in writing the heading rule changes the decision. This also explains why NBV is the only baseline that differs at all on the main suite: it is the only one that happens to use an action-consequence proxy, and as H3 shows, that proxy is not good enough.

4.2 Why next-best-view does not significantly beat history-only

Because geometric visibility and physical consequence differ systematically, and NBV only has access to the former. 43.1% of the actions geometry endorses fail physically. The paired difference against HistoryOnly is +0.042 with an interval of [0.00, 0.125] and p = 0.356 — not significant. On the paired-history worlds NBV scores 0/16, exactly like the other classical rules.

This is not an implementation issue. Placing the oracle at the executed pose recovers 97.9% agreement with physics. What fails is the prediction of where the action takes the robot.

4.3 The safety and horizon paradoxes are the same paradox

Both negative results have the same shape. A prediction carries real information — Oracle-S beats Oracle-V, and the horizon ablation shows that some terrains genuinely require multiple steps — and the decision rule fails to convert it. In the safety case this survives a targeted fix: halving the decision-band calibration error changes planning performance by exactly nothing. In the horizon case it survives increasing the depth: the chained score is optimistic in H, so more lookahead offers more ways to be wrong.

The common lesson is that prediction quality and decision usefulness are different properties. A head can rank well and still be unusable as a decision term; a model can be accurate one step ahead and still be unusable for multi-step ranking. Neither is fixed by more data, better calibration, or a longer horizon; both point at how consequences are combined into a decision.

4.4 Limitations

All results are from simulation, with a vendored third-party locomotion policy, on a single controller; no sim-to-real claim is made and the pose-consequence numbers should not be transferred across controllers. The oracle is a privileged upper bound and the search reference is incomplete: an earlier version of this work judged four terrains unsolvable, and beam search subsequently solved all four with two- to three-step sequences. That error — mistaking a planner's defect for a property of the benchmark — nearly put a spurious ceiling into the results, and the corrected solvability is 24/24 with an upper bound of 1.00 rather than 0.88.

The learned consequence model is evaluated with terrain held out, which is

generalisation within a mechanism family; cross-mechanism generalisation is not established here and an earlier single-step predictor scored below a naive baseline out of distribution. The target memory is a structured location representation reconstructed without noise, so belief-estimation failure is excluded by construction and claims about memory hold only for this representation. The action set is ten fixed primitives, and whether discretisation affects the H1 and H2 conclusions has not been tested. A binary fall label proved unusable — zero falls in 720 single-action rollouts — so safety is represented by boundary exit and a continuous stability margin instead, which means the learned safety head does not cover the event it is named after. Finally, the paired-world construction is available only in mechanisms that can express a directional aperture; Table 2 shows five mechanisms where this is structurally impossible, and the correct statement is that the construction is evaluated on the mechanisms that can carry it.

5 Conclusion

We asked what information and what action reasoning a robot needs in order to regain sight of a target that terrain has occluded. The answer arrived by elimination. The current observation is not sufficient: there exist worlds with pixel-identical observations whose optimal re-observation actions are opposite, and whose successful action sets are disjoint. A correct target memory is not sufficient either: there exist worlds identical in observation and in memory that still require different actions. Geometric visibility is not sufficient: 43.1% of the actions it endorses fail physically, while the same oracle at the executed pose agrees with physics 97.9% of the time.

What is missing is the action-conditioned physical consequence. Adding the candidate action to observation-plus-memory raises predictive AUC by 0.080, while adding the observation to the memory raises it by 0.002; a planner built on the resulting model

raises RSR@10 from 0.250 to 0.792 with no privileged information.

Two negative results bound how such a model may be used, and both are about the prediction-to-decision step rather than about prediction quality. A safety head that is accurate (AUC 0.900) and whose decision-band calibration can be halved by temperature scaling produces exactly zero planning gain, with no operating point beating simply not using it. A receding-horizon planner over the same model finds that deeper lookahead is not significantly better on any budget, costs nearly twice the motion, rescues two genuinely sequential terrains and breaks eight that one step solves.

The compact statement of the paper is therefore

O_t + M_t does not imply a^*, and the reason is the step in between:

a -> S_future -> V_future must be modelled

explicitly, must be action-conditioned, and must then be turned into a decision rather than a threshold.

6 Author Contributions

TO BE COMPLETED BY THE AUTHORS.

7 Funding

TO BE COMPLETED BY THE AUTHORS.

8 Data Availability Statement

The benchmark, baselines, consequence dataset and all result artefacts are generated by the code in the accompanying repository. Every figure and table in this manuscript is produced by a script from recorded JSON artefacts, so the numbers cannot drift from the figures.

9 Abbreviations

Non-standard abbreviations used in this manuscript, defined here as permitted immediately before the acknowledgments: AUC, area under the receiver operating characteristic curve; CI, confidence interval; ECE, expected calibration error; MAE, mean absolute error; MPC, model predictive control; NBV, next-best-view; OR, odds ratio; PR-AUC, area under the precision-recall curve; RSR, re-observation success rate.

10 Acknowledgments

TO BE COMPLETED BY THE AUTHORS. If generative artificial intelligence was used in preparing this manuscript, its name, version, model and source must be declared here and in the Methods.

11 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Table 1. Terrain-induced occlusion mechanisms. A mechanism is a point on an axis, not a member of a category: pit and basin are both spatial scale, at opposite ends of it.

Mechanism	Axis	Real-world analogue	Visibility effect
barrier	occlusion direction	walls, boulders, shelving	lateral occlusion
trench	elevation	ditches, trenches, ravines	edge occlusion
stairs_up	elevation	staircases, platforms	height-dependent
pit	spatial scale	potholes, craters	local depression
basin	spatial scale	valleys, broad depressions	large-scale occlusion
waves	visibility dynamics	dunes, rolling ground	repeated flips
perlin	terrain complexity	natural off-road ground	distributed occlusion
perlin_rough	terrain complexity	rubble, rock fields	irregular occlusion

Table 2. Mechanism support. A y-symmetric height field cannot express a directional aperture, so no amount of pair search will find a one-sided world in it.

Mechanism	$\lvert h(x,y)-h(x,-y)\rvert$	One-sided rate	Can carry the pair
stairs_up	0.0000 m	0.00	no (structurally)
trench	0.0000 m	0.00	no (structurally)
pit	0.0000 m	0.00	no (structurally)
waves	0.0000 m	0.00	no (structurally)

basin	0.0000 m	0.00	no (structurally)
barrier	0.0000 m	0.095	yes
perlin	0.2000 m	0.00	asymmetric, insufficient
perlin_rough	0.3000 m	0.00	asymmetric, insufficient

607

608 Table 3. Information ablation, leave-one-terrain-out, 2240 samples. The informative
609 comparison is O+M against O+M+a.

Inputs	ROC-AUC	PR-AUC	Brier	ECE	Pose MAE
O	0.759	0.360	0.117	0.076	0.436
M	0.852	0.639	0.077	0.039	0.461
O+M	0.854	0.600	0.084	0.056	0.425
O+M+a	0.934	0.748	0.062	0.022	0.148

610

611 Table 4. Pose-consequence error. The learned model wins on the components that decide
612 success and loses on height, where the analytic model predicts no change and changes are
613 small.

Component	Analytic (nominal)	Learned	Ratio
forward	0.829	0.152	5.5x
lateral	0.418	0.145	2.9x
height	0.090	0.149	analytic wins
yaw	0.538	0.147	3.7x
mean	0.469	0.148	3.2x

Table 5. Multi-step episode protocol, 24 terrains, identical budget and termination set for every method.

Method	RSR@1	RSR@3	RSR@5	RSR@10
six classical baselines	0.167	0.250	0.250	0.250
LearnedConsequence	0.708	0.708	0.792	0.792
Oracle (privileged)	0.833	0.875	0.875	0.875
OracleSearch (privileged)	0.750	1.000	1.000	1.000

Table 6. History ablation. Visibility prediction is flat, pose prediction improves and saturates.

H	AUC	Brier	Pose MAE
H0	0.930	0.063	0.177
H1	0.929	0.061	0.165
H2	0.926	0.063	0.157
H4	0.934	0.063	0.147
H8	0.934	0.062	0.148

Table 7. Action-horizon ablation. Intermediate states come from the learned model, never from ground-truth stepping; only the first action of each plan is executed.

H	RSR@1	RSR@3	RSR@5	RSR@10	Cost (m)
1	0.708	0.708	0.833	0.833	2.56

2	0.292	0.542	0.625	0.625	5.35
3	0.542	0.667	0.708	0.833	4.56
5	0.500	0.667	0.667	0.708	4.79

Table 8. Anchor-recovery baselines. Perfect localisation does not solve the task, and drift barely changes the outcome; the failures are physical, not positional. F denotes physical termination (fallen or out of bounds).

Drift	Mode	RSR@1	RSR@3	RSR@10	Failures
perfect	last-visible pose	0.125	0.125	0.167	B 14, F 6
perfect	backtrack	0.167	0.250	0.250	B 18
perfect	original ray	0.125	0.125	0.167	B 14, F 6
moderate	last-visible pose	0.125	0.125	0.167	B 14, F 6
moderate	backtrack	0.167	0.250	0.250	B 18
realistic	last-visible pose	0.125	0.125	0.167	B 13, F 7
realistic	backtrack	0.167	0.250	0.250	B 18

Table 9. Failure classification under protocol B, 24 terrains. B: a belief was present and a good action existed but was not chosen. F: the action was selected for a defensible reason and the physical outcome was not anticipated.

Planner	B action	F physical termination	other
HeadingLock/ HistoryOnly/	18	0	—

LostTargetSearch/ ObjectPermanence			
Reactive	17	1	—
NBV	16	1	C 1, D 1
LearnedConsequence	1	4	—
Oracle	3	0	—

Figure 1. Re-observation performance and where the discrimination comes from. (A) Re-observation success rate against action budget for every method, with Wilson 95% intervals; the five rule-based baselines coincide at every budget. (B) Success at the full budget, split by terrain group. Every classical method sits at 0.75 on the multi-terrain scenes and at 0.0 on the paired-history worlds; only the learned consequence planner and the privileged oracles leave zero.

Figure 2. Evidence established by the paired-world constructions. H1 pairs share a pixel-identical observation and opposite optimal actions with disjoint winning sets; H2 pairs additionally share the target memory and differ only in the optimal action.

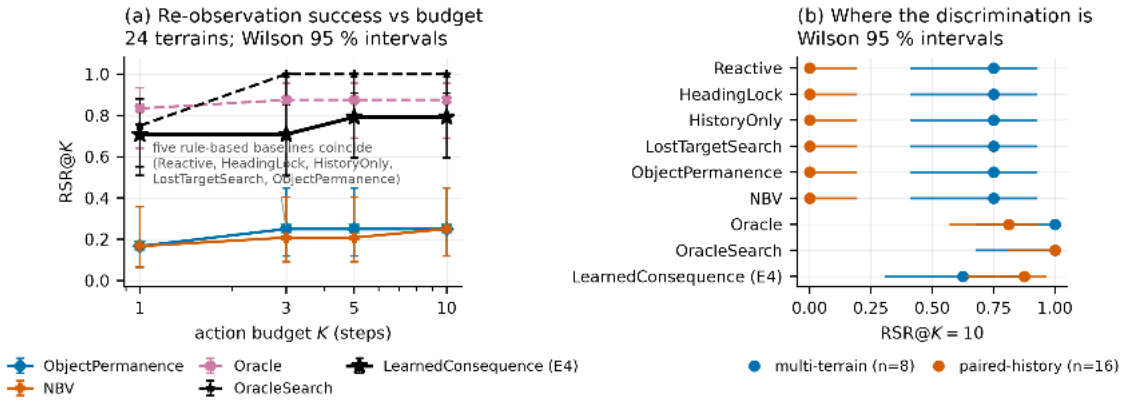
Figure 3. Geometric visibility against physical re-observation. (A) Confusion matrix over 240 (state, action) pairs; the outlined cell is geometry endorsing an action that fails physically. (B) Pose prediction error grouped by outcome, with the bootstrap contrast between the failing and agreeing cells.

Figure 4. History ablation. (A) Decision quality against history horizon, flat throughout. (B) Out-of-fold AUC, which saturates; the visibility prediction gains nothing from history while the pose prediction does.

Figure 5. Calibration against decision usefulness. (A) Overall expected calibration error compared with the error inside the decision band. (B) Success against the fraction of states at which the consequence term changed the action; no operating point beats not using it.

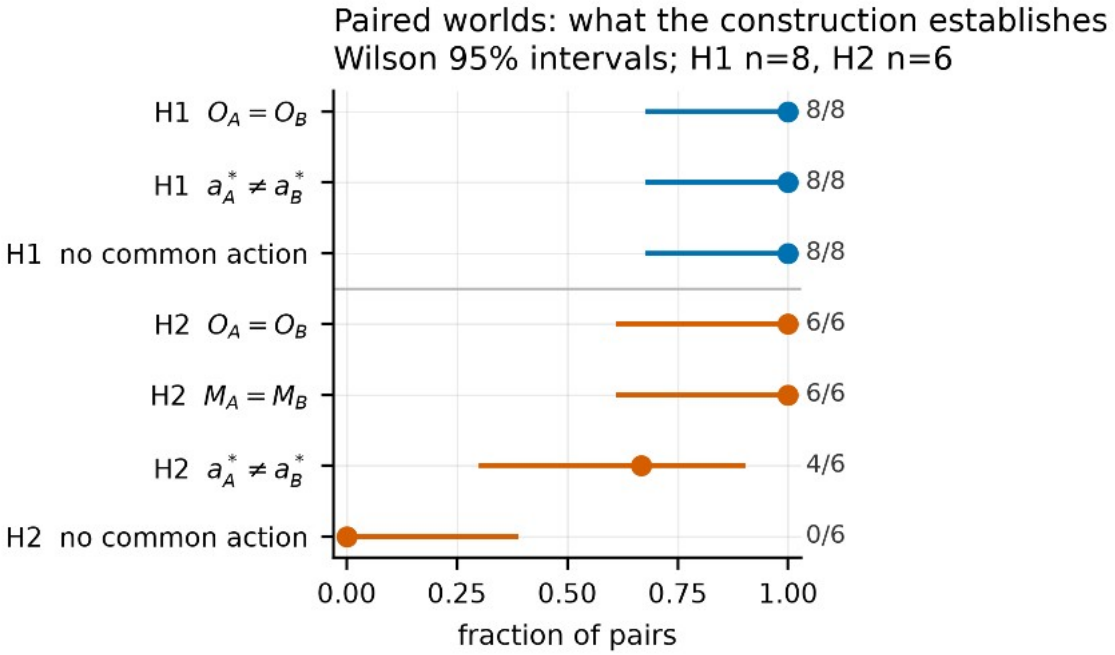
Figure 6. Action-horizon ablation. (A) Success by lookahead depth. (B) Mean action cost, which rises with depth. (C) The one-step pose error actually measured, which does not explain the ordering.

13 Figures



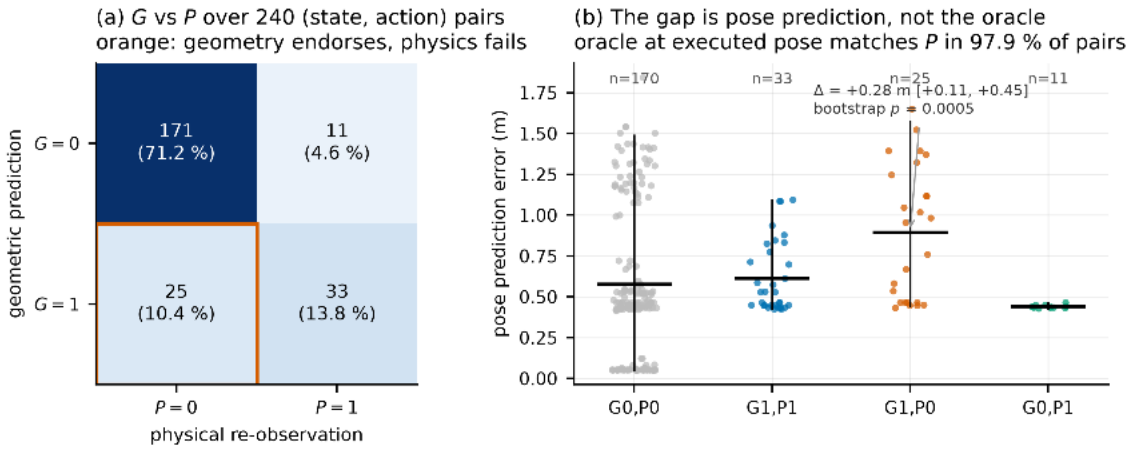
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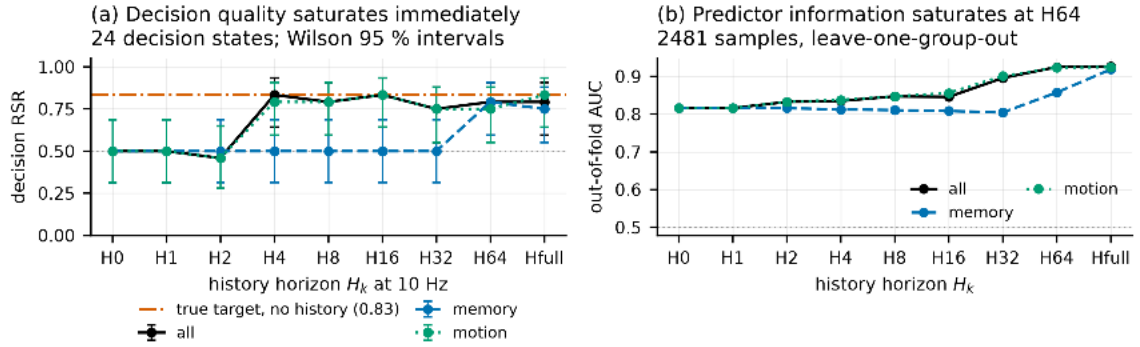


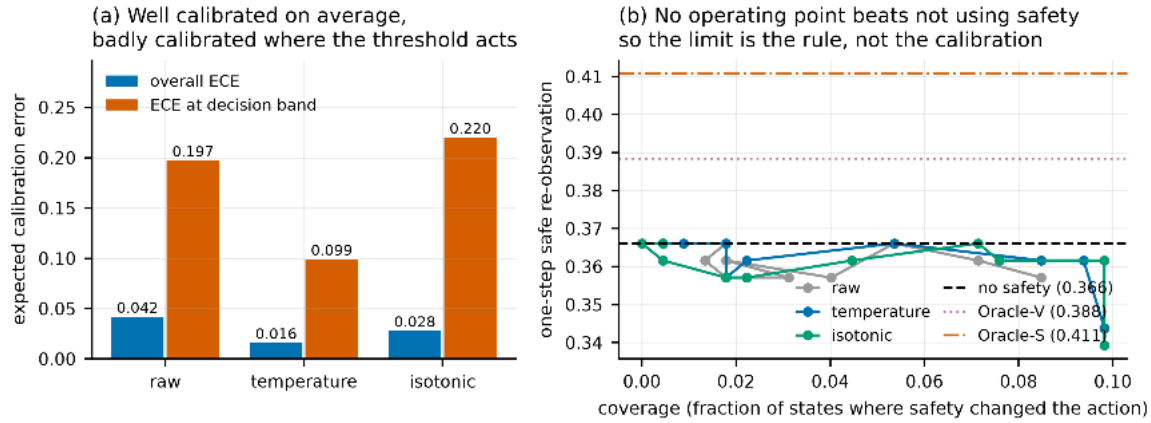
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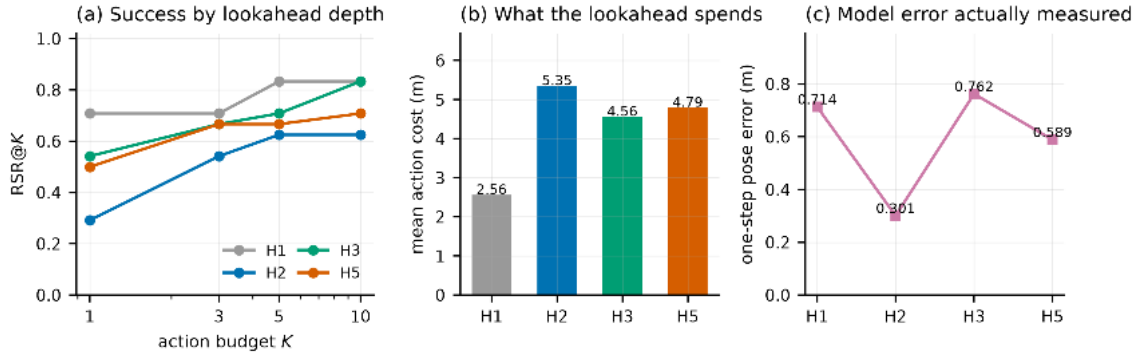
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(a) Decision quality saturates immediately
24 decision states; Wilson 95 % intervals
decision RSR
1.00
0.75
0.50
0.25
0.00
H0 H1 H2 H4 H8 H16 H32 H64 Hfull
history horizon H_k at 10 Hz
true target, no history (0.83)
all
memory
motion
(b) Predictor information saturates at H64
2481 samples, leave-one-group-out
out-of-fold AUC
0.9
0.8
0.7
0.6
0.5
H0 H1 H2 H4 H8 H16 H32 H64 Hfull
history horizon H_k
all
memory
motion


(a) Well calibrated on average,
badly calibrated where the threshold acts
overall ECE
ECE at decision band
expected calibration error
0.25
0.20
0.15
0.10
0.05
0.00
0.042
0.197
0.016
0.099
0.028
0.220
raw
temperature
isotonic
(b) No operating point beats not using safety
so the limit is the rule, not the calibration
one-step safe re-observation
0.41
0.40
0.39
0.38
0.37
0.36
0.35
0.34
raw
temperature
isotonic
no safety (0.366)
Oracle-V (0.388)
Oracle-S (0.411)
0.00 0.02 0.04 0.06 0.08 0.10
coverage (fraction of states where safety changed the action)


(a) Success by lookahead depth
RSR@K
1.0
0.8
0.6
0.4
0.2
0.0
H1
H2
H3
H5
1 3 5 10
action budget K
(b) What the lookahead spends
mean action cost (m)
6
5
4
3
2
1
0
2.56
5.35
4.56
4.79
H1 H2 H3 H5
(c) Model error actually measured
one-step pose error (m)
0.8
0.6
0.4
0.2
0.0
0.714
0.301
0.762
0.589
H1 H2 H3 H5